

CRESTRON MICROSOFT TEAMS ROOMS IMPLEMENTATION & DEPLOYMENT PLAN

Portfolio Sample — Demonstration Only

Sample Business	NorthStar Financial Services
Solution	Crestron Flex Microsoft Teams Rooms
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1. Executive Summary

This sample implementation and deployment plan describes the design, installation, configuration, testing, and production deployment of a **Crestron Flex Microsoft Teams Rooms (MTR)** solution for NorthStar Financial Services.

The solution is designed to provide a standardized video teleconferencing experience for executive meetings, business collaboration, training, and hybrid meetings.

The deployment includes a **Crestron UC Engine, Microsoft Teams Rooms software, room display/monitor, Teams Rooms touch controller, camera, microphone/audio components, network connectivity, and supporting infrastructure.**

The implementation follows a structured deployment lifecycle: **Requirements → Design → Site Assessment → Network Readiness → Hardware Installation → Configuration → Testing → UAT → Training → Cutover → Support.**

2. Sample Business

NorthStar Financial Services

Industry: Financial Services

Headquarters: Baltimore, Maryland

Regional Offices: Washington, DC; Philadelphia, PA; Richmond, VA

Employees: Approximately 750

NorthStar is implementing standardized Microsoft Teams Rooms technology to support collaboration between headquarters, regional offices, remote employees, and external business partners.

3. Business Objectives

- Standardize conference-room technology.
- Deploy Microsoft Teams Rooms.
- Improve audio and video meeting quality.
- Provide one-touch meeting joining.
- Support hybrid and remote collaboration.
- Provide a consistent user experience across meeting rooms.
- Reduce meeting-related technical support issues.
- Establish centralized device management.
- Provide documented installation and support procedures.

4. Proposed Crestron MTR Solution

Core Hardware

Crestron UC Engine

Crestron Teams Rooms touch controller

- Conference-room monitor/display

Crestron camera

- Microphone/audio solution

- UC Engine mounting/bracket assembly
- Required power supplies
- Required USB/network/video cabling
- Network switch connectivity

5. Sample Room Configuration

Standard Conference Room

Room Capacity: 8–10 people

Display: 1 × commercial-grade conference-room monitor

Compute: Crestron UC Engine

Room Control: Crestron Teams Rooms touch controller

Camera: Crestron-compatible collaboration camera

Audio: Conference-room microphone and speaker solution

Platform: Microsoft Teams Rooms

Network: Enterprise Ethernet

6. Crestron UC Engine

The UC Engine serves as the primary computing platform for the Microsoft Teams Rooms environment.

Configuration Activities

- Install the UC Engine.
- Connect required power.
- Connect network connectivity.
- Connect display output.
- Connect USB peripherals.
- Connect the Teams Rooms control interface.
- Verify device recognition.
- Apply required updates.
- Configure the room account.
- Validate Teams Rooms operation.

7. Conference-Room Monitor / Display

The room display will provide the visual interface for meetings.

Display Requirements

- Validate display resolution.
- Validate video input.
- Verify display power.
- Verify wake behavior.
- Verify sleep behavior.

- Verify HDMI/video connectivity.
- Verify display positioning and viewing angle.
- Validate cable management.

8. Microsoft Teams Rooms Configuration

The room will be configured as a dedicated Microsoft Teams Rooms resource.

Configuration Activities

- Establish the room/resource account.
- Configure Microsoft Teams Rooms licensing.
- Configure Microsoft 365 integration.
- Configure calendar integration.
- Configure room display name.
- Configure meeting policies.
- Configure device settings.
- Validate authentication.
- Validate network access.

9. Teams Rooms Touch Controller

The touch controller provides the primary room-user interface.

Users will be able to perform supported functions such as joining and starting meetings, ending meetings, adding participants, muting/unmuting, controlling supported camera functions, sharing content, and accessing meeting controls.

10. Network Readiness

Assessment

- Ethernet connectivity.
- Switch-port availability.
- VLAN assignment.
- DHCP.
- DNS.
- Internet connectivity.
- Firewall requirements.
- Microsoft 365 connectivity.
- Required service endpoints.
- Bandwidth.
- Latency.
- Packet loss.

- Jitter.
- **Readiness Criteria:** The room must have reliable network connectivity before production deployment.

11. Quality of Service

Where applicable, network QoS will be validated for real-time audio and video traffic.

Testing will include traffic classification, DSCP markings, network prioritization, WAN performance, latency, jitter, packet loss, and congestion behavior.

12. Physical Installation

Step 1 — Room Preparation

- Verify room readiness.
- Confirm power.
- Confirm network connectivity.
- Verify display location.
- Verify camera position.
- Verify microphone coverage.
- Confirm cable paths.

Step 2 — Hardware Installation

- Mount the display.
- Install the UC Engine.
- Install the camera.
- Install microphones/speakers.
- Install Teams Rooms touch controller.
- Connect required cables.
- Label equipment and cables.

Step 3 — Power-Up

- Power on display.
- Power on UC Engine.
- Power on peripherals.
- Verify device initialization.
- Confirm peripheral detection.

13. UC Engine Configuration

Hardware Validation

- UC Engine powers on.
- Network adapter is operational.

- Display connection is detected.
- USB peripherals are detected.
- Camera is detected.
- Audio devices are detected.

Software Validation

- Microsoft Teams Rooms application is operational.
- Required operating-system updates are applied.
- Required drivers are operational.
- Device configuration is correct.
- Room account authentication succeeds.

14. Microsoft Teams Rooms Integration

- Validate room account authentication, calendar synchronization, meeting scheduling, one-touch meeting join, internal meetings, external meetings, audio, video, content sharing, and meeting controls.

15. Camera and Video Testing

Testing will verify camera power, camera detection, camera image, camera framing, room coverage, video resolution, camera controls, and remote participant visibility.

- **Expected Result:** Remote participants should receive a clear and stable video image appropriate for the room configuration.

16. Audio Testing

Testing will include microphone pickup, speaker output, echo cancellation, mute/unmute, audio clarity, room coverage, remote participant audio, and local participant audio.

- **Expected Result:** Participants should be able to communicate clearly without unacceptable echo, distortion, or audio dropouts.

17. Content Sharing

- Content-sharing tests will validate wired presentation, supported wireless/content-sharing workflow, desktop sharing, presentation visibility, remote participant viewing, and local display behavior.

18. Interoperability Testing

- Test internal Teams users, remote Teams users, external meeting participants, different meeting-room locations, and supported collaboration workflows. Document any interoperability limitations or configuration dependencies.

19. Functional Test Plan

Test ID	Test	Expected Result
MTR-001	Power-on	System starts successfully
MTR-002	Network	UC Engine obtains network connectivity

Test ID	Test	Expected Result
MTR-003	Display	Monitor displays Teams Rooms interface
MTR-004	Touch controller	Controller operates correctly
MTR-005	Camera	Camera provides video
MTR-006	Microphone	Audio is transmitted
MTR-007	Speaker	Remote audio is received
MTR-008	Calendar	Scheduled meeting appears
MTR-009	Join meeting	One-touch join works
MTR-010	Content sharing	Shared content displays
MTR-011	External meeting	External participant connects
MTR-012	Mute	Mute function operates
MTR-013	End meeting	Meeting terminates correctly
MTR-014	Restart	System returns to operational state

20. User Acceptance Testing

Business representatives will perform UAT using normal meeting scenarios.

UAT Scenarios

- Schedule meeting.
- Enter conference room.
- Join meeting.
- Verify video.
- Verify audio.
- Share presentation.
- Add participant.
- Mute/unmute.
- End meeting.
- **Acceptance Criteria:** Critical tests pass; no unresolved critical defects remain; users can conduct meetings successfully; documentation is complete; support personnel are prepared.

21. Security Configuration

The implementation will follow the organization's security standards.

Security activities include dedicated room account, least-privilege administration, approved network access, secure authentication, device update management, configuration documentation, administrative access control, and monitoring/alerting.

- **Portfolio safety:** No passwords, authentication secrets, API keys, or confidential information will be included in portfolio documentation.

22. Device Management

Where applicable, Crestron's XiO Cloud capabilities may be used for provisioning, management, pairing, network management, and system alerts.

Management activities may include device inventory, device status, configuration management, alerts, provisioning, firmware/software lifecycle management, and troubleshooting information.

23. Training

End-User Training

- How to join meetings.
- How to start meetings.
- How to share content.
- How to mute/unmute.
- How to use camera controls.
- How to end meetings.
- Basic troubleshooting.

IT Support Training

- UC Engine troubleshooting.
- Teams Rooms troubleshooting.
- Network troubleshooting.
- Camera/audio troubleshooting.
- Display troubleshooting.
- Device management.
- Escalation procedures.

24. Production Cutover

Pre-Cutover Checklist

- Hardware installed.
- Network validated.
- UC Engine configured.
- Teams Rooms account configured.
- Display configured.
- Camera configured.
- Audio configured.

- Touch controller configured.

Testing completed.

- UAT approved.
- Documentation completed.

Cutover

- Place room into production.
- Validate calendar.
- Conduct production test meeting.
- Verify audio/video.
- Confirm content sharing.
- Confirm user controls.
- Monitor first production meetings.

25. Rollback Plan

If a critical production issue occurs: stop further deployment; isolate the affected room; document the failure; determine whether configuration remediation is possible; restore the previous room configuration if required; validate room availability; escalate unresolved hardware/software issues; and reschedule deployment after corrective action.

26. Support and Maintenance

Post-deployment support will include incident management, device monitoring, UC Engine troubleshooting, Teams Rooms troubleshooting, audio/video troubleshooting, network troubleshooting, software updates, hardware replacement, configuration management, and documentation updates.

27. Sample Deployment Schedule

Phase	Duration
Requirements	1 week
Room Assessment	1 week
Architecture & Design	1 week
Network Readiness	1 week
Hardware Preparation	1 week
Installation	1 week
Configuration	1 week
Functional Testing	1 week
UAT	1 week
Training	1 week

Phase	Duration
Production Cutover	1 week
Hypercare	1–2 weeks

Estimated deployment duration: 11–12 weeks.

28. Project Deliverables

Business requirements document

- Room assessment
- VTC architecture
- Hardware inventory
- Network readiness assessment
- Installation checklist
- UC Engine configuration
- Teams Rooms configuration
- Test plan
- Test results
- UAT results
- Training documentation
- Cutover checklist
- Support procedures
- Final deployment report

29. Final Acceptance

Crestron hardware is installed.

- UC Engine is operational.
- Microsoft Teams Rooms is configured.
- Room display is operational.
- Touch controller is operational.
- Camera and audio systems pass testing.
- Network connectivity is validated.
- Teams meetings operate successfully.
- Content sharing is validated.
- UAT is approved.
- Documentation is complete.

- Support ownership is established.

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